# COMPOSITE MATERIAL DATASHEETS CFRP AS4/8552-UD



## MATERIAL IDENTIFICATION

| | |
|---|---|
| Short informal name | AS4/8552-UD |
| Commercial name | Hexcel Hexply AS4/8552 RC34 AW194/196 |
| Specifications | Airbus: ABS5139C (AIMS05-01-001C grade B), ABS5347A (AIMS05-01-007)<br>CASA: Z-19.775/Z-19.780/Z-19.781 |
| Description | Carbon fiber unidirectional tape pre-impregnated in epoxy resin. 180ºC curing class structural material. |
| Version | 1.0 |

## REFERENCES

[1] TEA-T-NT-180037 Is. D “Residual Strength After Impact of CFRP Hexcel AS4/8552 UD-Tape Laminates”
[2] [NIAR] CAM-RP-2010-002 Rev. A “Hexcel 8552 AS4 Unidirectional Prepreg at 190 gsm & 35% RC Qualification Material Property Data Report”
[3] “Modelling and simulation methodology for unidirectional composite laminates in a Virtual Test Lab framework”, O. Falcó et al, 2018
[4] NT-A0-ECE-050 Ed. A "CRITERIO DE FALLO DE HERRAJES DEL BORDE DE SALIDA DEL CAJON LATERAL"

## PHYSICAL PROPERTIES

| | | |
|---|---|---|
| Cured Ply Thickness (*CPT*) | 0.184 | mm |
| Density | 1610 | kg/m$^3$ |
| Tg-onset (wet) | 159 | ºC |
| Max. Operational Limit (*MOL*) | 120 | ºC |

| Coefficients of Thermal Expansion (1/ºC) | Range | -55..20ºC | 20..70ºC | 70..120ºC |
|---|---|---|---|---|
| | $\alpha_1$ | -9.0E-07 | -1.6E-06 | -1.3E-06 |
| | $\alpha_2$ | 2.6E-05 | 2.9E-05 | 3.2E-05 |
| | $\alpha_3$ | 2.7E-05 | 2.8E-05 | 3.1E-05 |

## ELASTIC MODULI

| Property | $E_1$ | $E_2$ | $G_{12}$ | $\nu_{12}$ | $K_B$ |
|---|---|---|---|---|---|
| Units | MPa | MPa | MPa | - | - |
| Values | 133000 | 8800 | 4300 | 0.33 | 0.82 |

| Property | $E_3$ | $G_{13}$ | $G_{23}$ | $\nu_{13}$ | $\nu_{23}$ |
|---|---|---|---|---|---|
| Units | MPa | MPa | MPa | - | - |
| Values | 8800 | 4300 | 2700 | 0.33 | (*) |

*Remarks:*

- The $K_B$ is the buckling correction factor for moduli to apply to $E_1$, $E_2$ and $G_{12}$ in buckling stability analyses

(*) Result of the estimation of $\nu_{23}$ assuming transverse isotropy is too high (0.638) to be understood valid

## PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

| Pro-perty | Mean Values (MPa) | | | | | B-basis $K_B$ | *E-KDF* | | | | B-basis Values (MPa) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dry | Dry | Wet | Wet | Wet | | Dry | Wet | Wet | Wet | Dry | Dry | Wet | Wet | Wet |
| | -55ºC | RT | 70ºC | 90ºC | 120ºC | | -55ºC | 70ºC | 90ºC | 120ºC | -55ºC | RT | 70ºC | 90ºC | 120ºC |
| $F_{1t}$ | 2078 | 2135 | 1869 | 1825 | 1740 | 0.838 | 0.973 | 0.875 | 0.855 | 0.815 | 1742 | 1789 | 1566 | 1530 | 1458 |
| $F_{1c}$ | 1477 | 1546 | 1080 | 998.3 | 900.8 | 0.776 | 0.955 | 0.699 | 0.646 | 0.583 | 1146 | 1200 | 838.3 | 774.7 | 699 |
| $F_{2t}$ | 65 | 65 | 33.6 (1) | 33.6 | 24.7 | 0.768 | 1 | 0.517 | 0.517 | 0.380 | 49.9 | 49.9 | 25.8 | 25.8 | 19.0 |
| $F_{2c}$ | 366.8 | 283.8 | 190.8 | 165.2 | 138.5 | 0.805 | >1 | 0.672 | 0.582 | 0.488 | 295.3 | 228.5 | 153.6 | 133 | 111.5 |
| $F_{12}$ | 102.3 | 93.8 | 89.1 (1) | 89.1 | 57.4 | 0.722 | >1 | 0.951 | 0.951 | 0.612 | 73.9 | 67.7 | 64.4 | 64.4 | 41.5 |

(1) Experimental data for 70ºC/W conditions not available. Conservatively values at 90ºC/W conditions are adopted.

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

| Pro-perty | Mean Values (MPa) | | | | | B-basis $K_B$ | *E-KDF* | | | | B-basis Values (MPa) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dry | Dry | Wet | Wet | Wet | | Dry | Wet | Wet | Wet | Dry | Dry | Wet | Wet | Wet |
| | -55ºC | RT | 70ºC | 90ºC | 120ºC | | -55ºC | 70ºC | 90ºC | 120ºC | -55ºC | RT | 70ºC | 90ºC | 120ºC |
| $\varepsilon_{tall}$ | 11700 | 12700 | 12700 | 12700 | 12700 | 0.838 | 0.921 | 1 | 1 | 1 | 9800 | 10600 | 10600 | 10600 | 10600 |
| $\varepsilon_{call}$ | 11100 | 11600 | 8100 | 7500 | 6800 | 0.776 | 0.957 | 0.698 | 0.647 | 0.586 | 8600 | 9000 | 6300 | 5800 | 5300 |

## INTERLAMINAR PLY STRENGTHS

| Pro-perty | Mean Values (MPa) | | | | | B-basis $K_B$ | *E-KDF* | | | | B-basis Values (MPa) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dry | Dry | Wet | Wet | Wet | | Dry | Wet | Wet | Wet | Dry | Dry | Wet | Wet | Wet |
| | -55ºC | RT | 70ºC | 90ºC | 120ºC | | -55ºC | 70ºC | 90ºC | 120ºC | -55ºC | RT | 70ºC | 90ºC | 120ºC |
| $F_{13}$ | 141.8 | 110 | 78 | 67.5 | 56.2 | 0.89 | >1 | 0.71 | 0.613 | 0.511 | 126.2 | 97.9 | 69.5 | 60 | 50.1 |
| $F_{3t}$ | 58.0 | 54.4 | 40.5 | 36.0 | 29.3 | 0.52 [2] | >1 | 0.744[2] | 0.662[2] | 0.539 | 30.2 | 28.3 | 21.1 | 18.7 | 15.2 |

[2] CoV up to 25% in $F_{3t}$ batches. EKDFs interpolated within 0.95 at RT/Wet and value at 120ºC/Wet

| Pro-perty | Mean Values (J/m$^2$) | | | | | B-basis $K_B$ | *E-KDF* | | | | B-basis Values (J/m$^2$) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dry | Dry | Wet | Wet | Wet | | Dry | Wet | Wet | Wet | Dry | Dry | Wet | Wet | Wet |
| | -55ºC | RT | 70ºC | 90ºC | 120ºC | | -55ºC | 70ºC | 90ºC | 120ºC | -55ºC | RT | 70ºC | 90ºC | 120ºC |
| $G_{IC}$ | 300 [3] | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| $G_{IIC}$ | 540 [1]<br>870 [3] | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

[1] RT/Dry mean value reported at tests (n = 3, $S_d$ = 14.8 %); no more data available for fracture toughness.
(3) Values reported in research paper (Ref. [3])

## OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

| Pro-perty | Mean Values (MPa) | | | | | B-basis $K_B$ | *E-KDF* | | | | B-basis Values (MPa) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dry | Dry | Wet | Wet | Wet | | Dry | Wet | Wet | Wet | Dry | Dry | Wet | Wet | Wet |
| | -55ºC | RT | 70ºC | 90ºC | 120ºC | | -55ºC | 70ºC | 90ºC | 120ºC | -55ºC | RT | 70ºC | 90ºC | 120ºC |
| $OHT_{QI}$ | 317.3 | 338.6 | 338.6 | 338.6 | 338.6 | 0.9 | 0.937 | 1 | 1 | 1 | 285.6 | 304.7 | 304.7 | 304.7 | 304.7 |
| $OHC_{QI}$ | - | 325.6 | - | - | 233.1 | 0.9 | - | - | - | 0.716 | - | 293.1 | - | - | 209.8 |
| $FHT_{QI}$ | 362.9 | 334.5 | 334.5 | 334.5 | 334.5 | 0.838 | >1 | 1 | 1 | 1 | 304.1 | 280.3 | 280.3 | 280.3 | 280.3 |
| $FHC_{QI}$ | - | 502.4 | - | - | 386.5 | 0.773 | - | - | - | 0.769 | - | 388.5 | - | - | 298.8 |
| $F_{BR-QI}$ | - | 999 | 700.2 | 656.5 | 575.5 | 0.895 | - | 0.701 | 0.657 | 0.576 | - | 893.6 | 626.3 | 587.2 | 514.8 |

## COMPOSITE DAMAGE TOLERANCE

- Residual strength after transverse impact

  The following TAI, CAI model is valid for laminates following the standard design rules (symmetric, balanced, with minimum 10% of plies at angles 0º, 90º, 45º and -45º).

| Property | B-basis Value for all conditions |
|---|---|
| *TAI* (με) | 5100 |

$E/t^{\alpha} = A \cdot (1 - e^{-B \cdot d0/t})$

| Parameter | Design Value |
|---|---|
| *A* (J/mm²) | 4.6 |
| *B* | 1.4 |
| *α* | 2.0 |

*Approximation model type: CDTM-T07*

$CAI = K_B \cdot EKDF \cdot CAI_{\infty\text{-}QI} \cdot (3 / K_{TOH})^{Cx} \cdot [\ 1 + A / e^{Be \cdot (E/t^{\wedge}\alpha)}\ ]$

| Parameter | Mean Value RT/Dry | B-basis $K_B$ | E-KDF | | |
|---|---|---|---|---|---|
| | | | 70ºC/W | 90ºC/W | 120ºC/W |
| $CAI_{\infty\text{-}QI}$ (με) | 3160 | 0.87 | 0.90 | - | 0.78 |
| A (με) | 0.67 | | | | |
| $B_e$ (mm²/J) | 1.11 | | | | |
| $C_x$ | 1.0 | | | | |
| *α* | 2.0 | | | | |

…where:

- *E* is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- *t* the laminate thickness
- $d_0$ the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- $K_{TOH}$ is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{\ 2 \cdot [(E_x/E_y)^{0.5} - v_{xy}] + E_x/G_{xy}\ \}^{0.5}$$

(*) CAI shall be calculated not only for the 0º direction (X parallel to material 0º reference), but also for the rest of ply orientations, i.e. the 45º and 90º directions, by rotating the X axis and re-evaluating previous expressions with the new values of $E_x$, $E_y$, $G_{xy}$ and $v_{xy}$ . In a general case CAI strength shall be expressed as 3 scalar values with the alowable strains applicable to plies at 0º ($CAI_0$), at ±45º ($CAI_{45}$) and at 90º ($CAI_{90}$).

- Residual strength after transverse BVID near lightening holes

| Property | B-basis value up to 70ºC/W |
|---|---|
| $CAI_{hole}$ (με) | 7200 |

…to be compared with the peak tangential strain calculated at the edge of a hole in a panel made of this material. Source: legacy data (see Ref. [3]).

- Residual strength after edge impact

No design data for *CAIE* still available